

Team 1

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Engineering Service Learning

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Element B:

[Prior Attempt 1:](#)



This solution works as a lock to hold a door closed¹. It can be used to push against the door using the floor as support or to push against a sliding door using the door frame as support. This solution will not work since it is set up on the inside of the door and our solution must be able to be used from the outside. This solution also only works on inward opening doors, so it will not work on all the doors that may be encountered². This solution is too large to be carried around easily and our solution must fit in the front pocket of a pair of pants. This solution does have the benefit of being manageable with one hand³.

Prior attempt 2:



This solution attaches to the door handle mechanism and acts as a physical barrier when attempting to open the door. This solution only works standard doors that fully seal/fill the frame and have a standard door handle mechanism. Many bathroom doors do not fully reach the frame so this solution would not work. This solution cannot be properly utilised with one hand and, as per our interview, our solution must be operable with a single hand. This is useful because it could fit in the operator's pocket.

Prior attempt 3:



This solution works in a similar way to Prior Attempt 2, but with a different method to activate the lock. This solution also only works with standard doors, so the same problem of many bathroom doors not fully filling the frame is present here as well. This solution can be operated quickly and with one hand, so we could use this for ideas on how to design our solution.

[Prior attempt 4](#)



This solution uses friction to prevent the door from being moved. This solution would not work on all doors since it requires the door to be close to the floor. This is very uncommon in bathroom stall doors. This solution cannot be operated quickly (see video in amazon link). It must also be attached on the inside of the door and our solution has to work from the outside. One benefit of this solution is that it's manageable with one hand.

Prior attempt 5

This works by holding on to the slot in the door frame for the door handle mechanism and then uses stoppers on the inside of the door to prevent it from moving. This solution does not work since it must be attached on the inside of the door but our solution must work from the outside. It also requires a traditional door handle system, which are not common for bathroom stall doors. It is an approach that we can take ideas from because of its sturdiness and the ability to tighten/loosen with one hand¹.

Works Cited

1. Fallahi, Ali, and Ali Jadidian. "The Effect of Hand Dimensions, Hand Shape and Some Anthropometric Characteristics on Handgrip Strength in Male Grip Athletes and Non-Athletes." *Journal of Human Kinetics*, vol. 29, no. 1, 1 Sept. 2011, pp. 151–159, <https://doi.org/10.2478/v10078-011-0049-2>.
2. Transportation Security Administration. "What Can I Bring?" Transportation Security Administration, 2019, www.tsa.gov/travel/security-screening/whatcanibring/all.
3. "U.S. Access Board - ADA Accessibility Standards." *Www.access-Board.gov*, www.access-board.gov/ada/#ada-309_4.
4. Dale, Timothy. "How to Lock a Door without a Lock: 10 Brilliant Solutions." *Bob Vila*, 23 Dec. 2022, www.bobvila.com/articles/how-to-lock-a-door-without-a-lock/.